

Towards Solver-Based Labelling of Player Tactics in Logic Puzzles

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Abstract

In order to model and assess player knowledge and understanding of game concepts, it is necessary to identify when players apply those concepts. This can be difficult in games with complex logical reasoning tactics, as there may not be in-game actions clearly demonstrating the reasoning the player carried out. To explore this challenge, we examined logic grid puzzles, in which players use textual clues to reason about relationships among entities on a grid. In this work, we developed an approach to identifying the tactics that players applied during gameplay, by finding the next logical moves according to a human-reasoning-based puzzle solver. We analysed unlabelled puzzle playing data collected in an online user study. We found that our approach can label tactics for 77 percent of successful players' correct moves, and players show a moderate tendency to use similar tactics to solve a given puzzle. However, we do not have ground truth data to assess labelling accuracy. This work is a first step in detecting player reasoning during logical puzzle solving.

1 Introduction

Game designers often need to model players' knowledge of game concepts to design an effective level progression, especially for dynamic difficulty adjustment, which aims to dynamically serve players levels that align with their current skill level (Kantharaju et al. 2018; Sarkar and Cooper 2020). In particular, when game designers know what concepts players are struggling with, they can provide support that enhances learning, rather than relying on hints that offer direct solutions (Ali and Thue 2025).

Traditionally, knowledge tracing systems estimate player knowledge based on completion or failure of a task testing one or more "knowledge concepts" (Shen et al. 2025). However, a simple completion metric does not always accurately capture player knowledge. A player taking random actions may still be successful, and players often interweave skills as they focus on different aspects of a task (Kantharaju et al. 2018; Hooshyar, Huang, and Yang 2022). Research into knowledge tracing for games frequently addresses the question of whether players are behaving randomly or strategically while assuming that the skills needed to complete game tasks are known (Kantharaju et al. 2018; Hooshyar,

Huang, and Yang 2022). However, in games with complex reasoning (such as deductive logic puzzle games), players can arrive at a correct solution through multiple combinations of "tactics" (Horn et al. 2016), which are discrete logical inferences used in cognitive reasoning about a puzzle solution. Therefore, it becomes necessary to follow player reasoning through the individual moves that they make.

To identify logical reasoning tactics in deductive puzzle games, we developed a simple tactic-labelling system that compares player moves to a set of suggested moves (with associated tactics) provided by a designer-specified "oracle". The oracle outputs logical moves that could be made given the current state, each associated with the logical tactic used to make that move. We tested the capability of this method on the domain of logic grid puzzles, a kind of logic puzzle in which players identify relationships between entities on a grid by resolving a set of constraints given as natural language "clues". We modified an automated logic grid puzzle solver to serve as an oracle that suggests moves based on the current puzzle state. To disambiguate between a variety of tactics associated with the same move, we label the move with the tactic that is least complex, according to a designer-specified ordering.

We tested our logic grid puzzle implementation on unlabelled user data from an online study. Our tactic-labelling system was able to label 77% of correct user moves on successful attempts, and solutions to the same puzzle tend to be labelled with similar tactics. Comparing the results of online players (successful and conceding), an expert player, and randomized agents, we find that the percentage of moves labelled relates to player skill.

This method represents a first step toward improved detection of player reasoning in puzzle games. The contributions of this work are as follows:

1. A generalizable logical tactic-labelling method for player actions in deductive puzzle games.
2. Demonstration of the method applied to the domain of logic grid puzzles.
3. Analysis of the approach applied to user data showing that it may be able to differentiate players by skill.

2 Related Work

2.1 Knowledge Tracing

Knowledge tracing is a wide field of work that attempts to describe learners’ success in learning concepts or skills (Shen et al. 2025). A common form of knowledge tracing is Bayesian Knowledge Tracing (BKT), which models knowledge concepts as latent (hidden) nodes estimated from student performance (Šarić Grgić, Grubišić, and Gašpar 2024). Many embellishments have been proposed to the vanilla model over the years, including incorporating learner-specific parameters and modelling forgetting, or knowledge decay (Šarić Grgić, Grubišić, and Gašpar 2024). Generally, knowledge tracing assumes that the concepts used are known, and learners’ knowledge is estimated according to their success on different exercises with pre-assigned concepts.

2.2 Skill Chains

In games, knowledge tracing is commonly represented using “skill chains”. Cook (2007) introduced the concept of skill chains: dependency graphs composed of “skill atoms” representing player skills that build on each other. Skill chains have been used as the basis for both static (Linehan et al. 2014; Horn et al. 2017) and dynamic (Sarkar and Cooper 2020) level progressions. Often, as with general knowledge tracing, player skill is estimated mainly by the levels they successfully solve, rather than by player behaviour during a level.

Some researchers have attempted to identify player skills during gameplay through analysis relying heavily on time-consuming manual analysis of play data. Horn et al. (2016) used hierarchical clustering and qualitative analysis of think-aloud data to identify different player strategies for solving the puzzle game *GrACE*. Later, Horn, Cooper, and Deterding (2017) used think-aloud and stimulated recall data to generate a skill chain for the human computation game *Paradox*. Louis Lee, Goodwin, and Johnson-Laird (2008) identified “tactics” used by novice Sudoku players by having them explain each move they made. Kantharaju et al. (2018) used machine learning, with think-aloud data as ground truth, to identify when players attempt to employ particular skills and how successful they are.

Only a couple works used purely automated gameplay analysis to track player skills. Baek, Park, and Kim (2023) used MCTS to generate tutorial levels designed to train players towards a particular play style in a match-three game by automatically tracking their skill use during the level, although they do not provide details of how skill use was determined. Aytemiz et al. (2018) offer a tutorial and hint system inspired by Dan Cook’s skill atoms that works by attaching skills to game objects that serve as “detectors”. While our method is similar in that it relies on designer-specified-skills, our method is geared toward logical tactics in puzzle games rather than the knowledge of mechanics in action-adventure games. In addition, their method requires designers to manually add skill detectors to each level, while our method takes advantage of an automated solver to detect tactics.

2.3 Puzzle Tactics

A number of papers have focused on the “tactics” that humans use to solve logic puzzles, such as Sudoku and Nonograms, often to approximate their difficulty (Louis Lee, Goodwin, and Johnson-Laird 2008). Logic puzzle tactics are puzzle solving strategies or shortcuts typically compiled by puzzle experts (Hanson 2006). They go by various other names, including “strategies” (Horn et al. 2017), “inference rules” (Mahmoud and Sturtevant 2024), or just “techniques” (Pelánek 2014). Researchers seldom use expert-compiled tactics directly, preferring to create computational models of difficulty (Louis Lee, Goodwin, and Johnson-Laird 2008; Benson 2017). Butler, Torlak, and Popović (2017) describe a method to generate human-understandable puzzle-solving tactics from constraint-solving methods and automatically generated solution traces.

A common use of tactics is to describe puzzle solving difficulty. Occasionally, difficulty is estimated according to how many times a given tactic is used (Batenburg and Kusters 2012). More commonly, tactics are ranked according to subjective or programmatic complexity, and a puzzle’s difficulty is estimated by its most complex tactic (Batenburg et al. 2009; Oranchak 2010). On the other hand, Pelánek (2014) asserts that advanced logical tactics represent a form of search using simple tactics. Similarly, Browne (2013) estimates puzzle difficulty according to the depth of search required to solve the puzzle using simple constraint satisfaction tactics and limited puzzle-specific strategies.

Others have researched the process of teaching specific puzzle solving tactics. A common strategy is to generate puzzle levels that encourage (Mahmoud and Sturtevant 2024; Jemmali, Seif El-Nasr, and Cooper 2022), or require (Cooper and Bazzaz 2024; Smith, Butler, and Popovic 2013), the use of a particular tactic or move. As with knowledge tracing methods, in these cases, players’ use of tactics can only be estimated by their ability to complete levels that enforce the use of those tactics. Our method is intended to identify player use of tactics during gameplay, which in the future can be used in combination with these techniques to guide and assess player knowledge of tactics, including tactics that are not enforced by a particular level’s design.

3 Tactic Labelling

We have developed a novel method for labelling player actions with the tactics players may be using as they solve a deductive logic puzzle.

3.1 Terminology

Deductive Logic Puzzle A puzzle that may be solved by a series of constraint applications not requiring extensive guesswork or lookahead (Browne 2013).

Tactic A logical deduction rule that may be used to make a change to the puzzle state.

Tactic Ordering A list of tactics ordered in terms of complexity. Less complex tactics are preferred to more complex tactics when assigning tactics to user moves.

Algorithm 1: Tactic Labeller

Input:

(s_i, m_i) – the current state and the next user move
 O – the oracle function $O : S \rightarrow M \times T$; where m_t is the move applying tactic t to s_i
 T – the tactic ordering

Output: t_i – the tactic label for m_i

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let  $t_i$  be null
 $M_i = O(s_i)$ 
for  $(m_t, t) \in M_i$  do
  if  $m_t == m_i$  and  $t < t_i$  in  $T$  then
    set  $t_i = t$ 
  end if
end for
  
```

Suggested Move A logical state change, with associated tactic.

Oracle A source that can be queried for suggested moves, given the current state.

3.2 Labelling Process

Given the current puzzle state and the next user move, our system uses the oracle to attempt to label the move with a corresponding tactic using the following process (represented by Algorithm 1).

First, we query the oracle for a set of suggested moves given the puzzle state. The oracle returns all valid moves associated with a known tactic. We attempt to find one or more suggested moves that make the same state change as the user move. If any are found, we label the user move with the tactic that is lowest in the tactic ordering. Otherwise, we record that there is no associated tactic. The process of generating suggested moves and ordering tactics is necessarily domain-specific, so to demonstrate the potential of this method, we implemented these steps for a case study using the domain of logic grid puzzles.

To better represent the player’s knowledge, the oracle does not incorporate knowledge of the puzzle solution. That is to say, the oracle suggests moves based only on the puzzle state and the tactics. When the current state agrees with the solution, the oracle never suggests an invalid move. However, *the purpose of the oracle is not to provide correct moves*, as would be used in a hint system, but to give suggestions to the labelling system of *tactics the player might use*. Thus, if the current state is wrong (the player made a mistake), the oracle might suggest moves that do not align with the solution but are consistent with the current state. Given that the oracle considers each tactic independently, it may even suggest moves that contradict each other if the board state is incorrect. This allows us to label player moves according to what may be the current beliefs of the player, rather than according to the solution.

4 Domain Case Study: Logic Grid Puzzles

To demonstrate our tactic labelling method, we choose to use the domain of *logic grid puzzles*, a type of puzzle also

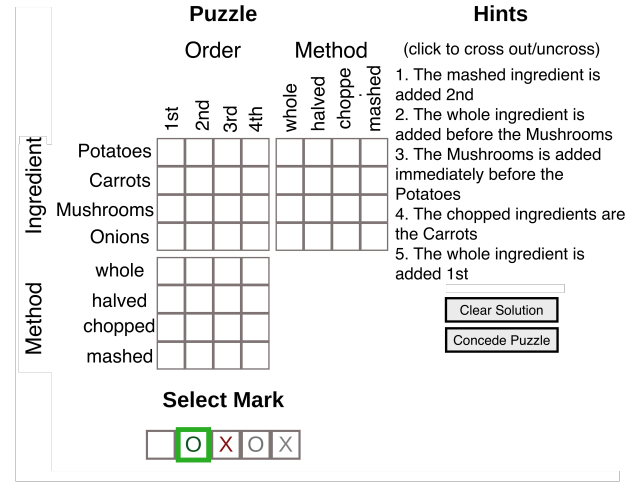


Figure 1: An example of a logic grid puzzle, in the interface used by online users. Here, the clues are referred to as “Hints”. “O” is used to indicate a relationship, while “X” eliminates one. The red and green marks are confident, while the grey marks are tentative.

known as “Einstein” or “Zebra” logic puzzles¹. In these puzzles, players must link the *entities* belonging to different *categories* using a series of natural language *clue constraints* giving information about which entities are related in the form of constraints. Typically, there are also two *global constraints* on entity relationships. First, relationships are one-to-one; that is, every entity is related to exactly one entity in every other category. Second, relationships are transitive, such that if entity A is related to entity B and B is related to entity C, then A must also be related to C. Typically, to reduce mental bookkeeping (keeping information “in one’s head”), players are provided with a grid in which each unique pair of categories has a sub-grid. An example logic grid puzzle is provided in Figure 1.

Logic grid puzzles are interesting from a tactic labelling perspective because the variety of possible clues lends itself to complex logical reasoning, and the use of a grid allows users’ thought processes to be partially observed. Logic grid puzzles are also convenient, as we previously developed a solver hand-authored based on human reasoning processes (Shyne, Facey, and Cooper 2024a,b, 2025).

4.1 Terminology

Here we describe some of the terminology we use when discussing logic grid puzzles and their solving.

Category A group of related entities. Each entity is related to exactly one entity in every other category.

Clue Constraint A constraint pertaining to specific entities and given to the player in a natural language “clue”.

Global Constraint A constraint pertaining to all entity relationships (they are one-to-one and transitive).

¹Data and code are publicly available at https://github.com/kaylah-fae/logic_puzzles/releases/tag/tv1.0.0

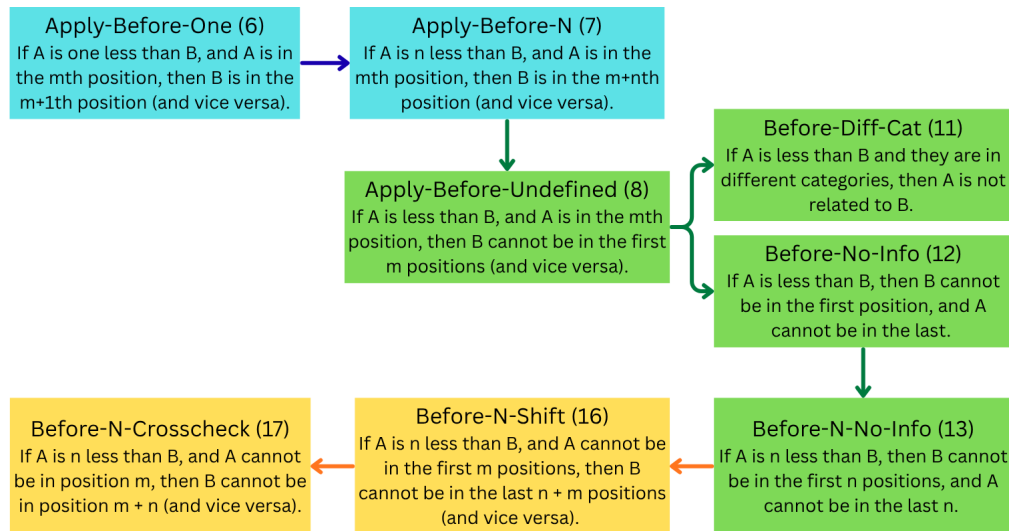


Figure 2: Tactic dependencies for “before” clues, with their position in the tactic ordering in parentheses. Blue tactics are direct applications of the clues based on the grid state. Green tactics are derived from the clues and can be applied to a blank grid. Yellow tactics require that some information be marked on the grid.

Grid The entire puzzle grid.

Sub-grid A single category pair on the grid.

Cell One location in the grid representing the relationship between two entities.

Mark A character representing the confirmed or negated relationship between two entities. Marks may be positive or elimination, and confident or tentative.

Positive Mark An “O” mark representing the existence of a relationship between two entities.

Elimination Mark An “X” mark representing the negation of a relationship between two entities.

Confident Mark A mark that is confident.

Tentative Mark A mark that is uncertain. Tentative marks may represent guesses or the enumeration of possible relationships.

4.2 Logic Grid Puzzle Oracle Tactics

To build an oracle for this work, we modified our hand-authored logic grid puzzle solver. The automated solver uses an iterative process, making logical deductions for each clue in turn. Although tactics are not labelled explicitly, they are implicitly present in the deductions made, as the solver is intended to mimic human puzzle solving. Figure 3 serves as a running example, demonstrating a number of logic grid puzzle tactics for a logic grid puzzle with some information already filled out and the clue “Col. Mustard arrived 2 hours before Ms. Scarlet”.

For this work, we manually identified the tactics that the solver uses. We also designed a tactic ordering from least to most complex, based on our expert intuition. The simplest tactics are straightforward applications of the clues, while the most complex tactics require advanced logical deduction. On the lower end, the Cross-Out and Opening tactics are straightforward applications of the global constraint

that relationships are one-to-one. As seen in Figure 3c, since Prof. Plum arrived at 5pm, then he did not arrive at any other time (Cross-Out). Inversely, since Col. Mustard did not arrive at 5pm, 7pm, or 8pm, then he must have arrived at 6pm (Figure 3f; Opening).

Tactics in the ordering often derive from each other. For example, for the clue “Col. Mustard arrived 2 hours before Ms. Scarlet”, a number of tactics can be applied. The set of dependencies for “before” clues is illustrated by Figure 2. At the least complex, by the Apply-Before-N tactic, if Col. Mustard arrived at 6pm, then Ms. Scarlet must have arrived at 8pm (Figure 3h).

A tactic is always later in the tactic ordering than another tactic if it is a more complex form of similar logic. For example, Before-N-No-Info is a more complex tactic for “before” clues that can be used on an empty grid. Even if it is unknown at what time any suspect arrived, it can still be deduced that Col. Mustard could not have arrived in the last two hours, and Ms. Scarlet could not have arrived in the first two hours (Figure 3b). Where tactics are not obviously related, we order the tactics using our expert intuition for which tactics are more difficult to deduce and employ.

4.3 Logic Grid Puzzle Oracle Suggested Moves

After identifying and labelling the solver’s tactics, we modified it to serve as an oracle. Where the original solver applies all tactics for a given clue at once, our oracle suggests moves for each tactic independently.

Given a puzzle’s clues and current state, the oracle iteratively loops over the clues. For each clue, it attempts to apply all tactics independently. For each tactic applied, the oracle returns a suggested move including the state change, the tactic used, and the relevant clue. Because the oracle’s suggestions are independent, it is possible that it suggests the same mark for multiple different clues or tactics.

Handling Uncertainty The solver was designed to make only marks it is confident in and cannot solve puzzles that require guessing. However, players of pen-and-paper logic puzzles often do employ tentative or “pencil” marks to denote information they are uncertain of or to mark possibilities (Crook 2009).

In logic grid puzzles, we have found that users make tentative marks for two main reasons. Some marks appear to be guesses (such as marking one specific time Col. Mustard could have arrived given that he arrived 2 hours before Ms. Scarlet). Others seem to denote possibilities (such as marking all of the times Col. Mustard could have arrived). The original solver accounted for possibilities by making elimination marks in all the non-possibilities. To attempt to capture the user behaviour of making tentative positive marks for possible relationships, the oracle includes suggested moves denoting tentative positive marks for remaining possibilities (as shown in Figure 3b). Tentative marks and elimination marks are separated into different suggested moves for the same tactic.

4.4 Logic Grid Puzzle Labelling

Logic grid puzzle tactic labelling in general proceeds as described in Section 3, with two major additions. First, we augment the suggested moves with additional moves to capture the likelihood that the user is holding some information “in their head” (mental bookkeeping). Second, we map multiple user moves to a single suggested move where possible. Figure 3 demonstrates the process with a running example.

Handling Mental Bookkeeping The solver is designed to ensure that puzzles don’t require backtracking, as humans have limited capacity for thinking ahead while puzzle solving (Pelánek 2014). and often prefer not to guess when solving deductive puzzles (Butler, Torlak, and Popović 2017). However, we have observed that users appear to use some mental bookkeeping while solving logic grid puzzles. In particular, users do not always mark eliminations, preferring to solve a puzzle using primarily positive marks. Therefore, to partially account for this behaviour, we augment the oracle’s suggested moves in two ways. Algorithm 2 demonstrates this process.

First, we apply all suggested Cross-Out moves to the original puzzle state and take the suggested moves for that state (Figure 3c). This reflects the fact that if a user has made some positive marks, they may be taking the one-to-one relationships rule into account mentally.

Second, we check each suggested move’s state change (including those added for the Cross-Outs), and if the state change consists purely of elimination marks, then we apply that move to the puzzle state and request suggested moves for the ensuing state (Figure 3b; 3e). We keep any resulting moves using the Opening tactic (Figure 3f) and label them with the original suggested tactic. This reflects the fact that a user may mentally apply the elimination marks for a move and physically only add the resulting Opening positive mark to the puzzle.

Adding moves through the augmentation process does not eliminate the possibility of labelling the intermediary marks

Algorithm 2: Augmented Suggested Moves

Input:

s_i – the current puzzle state

O – the oracle function $O : S \rightarrow M \times T$; where m_t is the move applying tactic t to s_i

Output: M_i – the augmented suggested moves for s_i

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 $M_i = O(s_i)$ 
Augment with post-Cross-Out moves
for  $(m_t, t) \in M_i$  do
  if  $t == \text{Cross-Out}$  then
     $s_{co} = s_i + m_t$ 
     $M_{co} = O(s_{co})$ 
     $M_i = M_i \cup M_{co}$ 
  end if
end for
Augment with post-elimination Opening moves
for  $(m_t, t) \in M_i$  do
  if  $m_t$  consists only of elimination marks then
     $s_t = s_i + m_t$ 
     $M_o = O(s_t)$ 
    for  $(m_o, t_o) \in M_o$  do
      if  $t_o == \text{Opening}$  then
        add  $(m_o, t)$  to  $M_i$ 
      end if
    end for
  end if
end for

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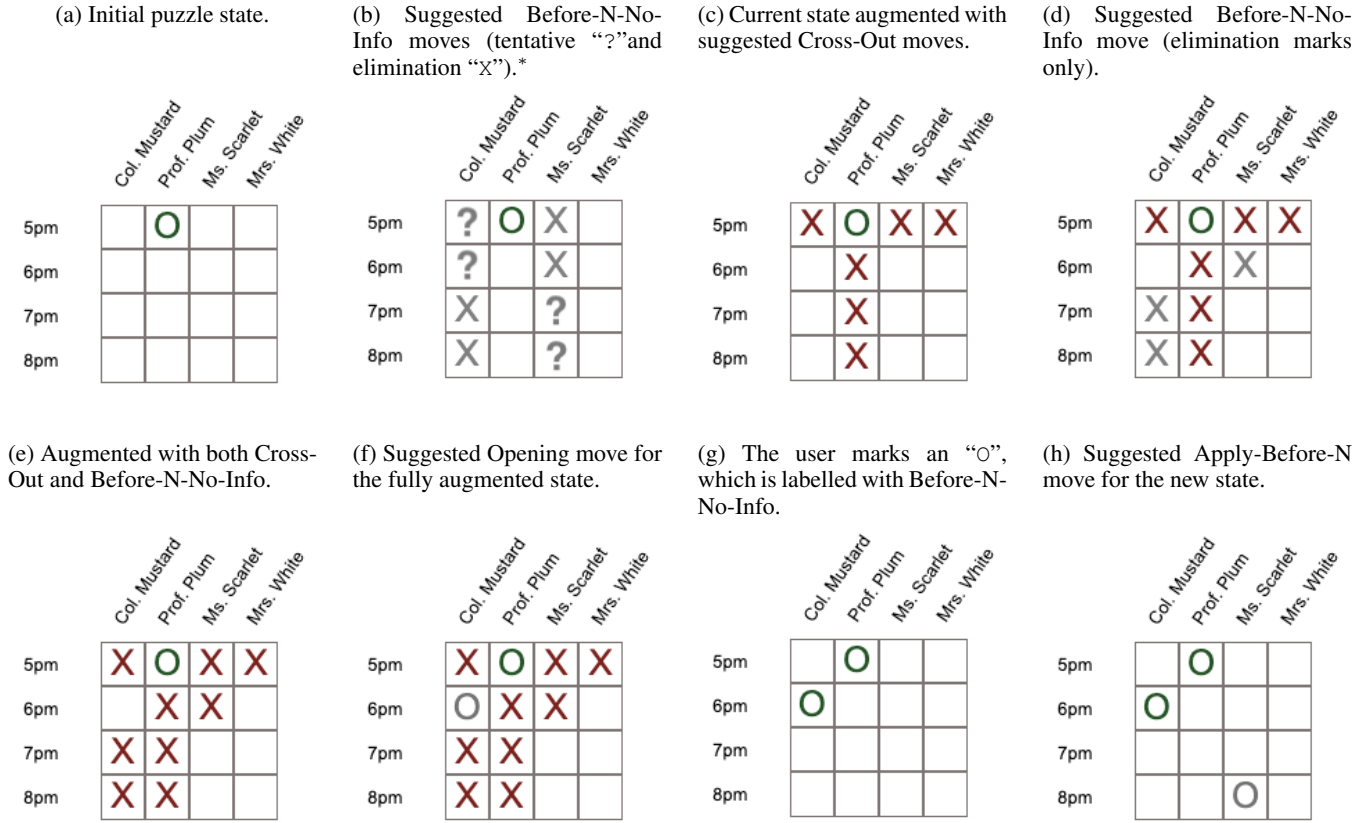
captured by the augmented grid. A player can either make these marks explicitly (and have intermediary solution steps labelled) or implicitly (through using a tactic found for the augmented grid).

Handling Granular User Data While the oracle can suggest moves that make multiple marks at once, users are limited to one mark at a time. Each suggested move making n marks is compared to the next n user moves. If the user moves from t to $t + m - 1$; $m \leq n$ each make a mark that is also made by the suggested move, then all m user moves are labelled with that suggested move’s tactic, provided that $m \geq 2$ (if only the first user move matches, then it is not labelled unless n is 1). Longer runs of matching moves are preferred, and as described above, if multiple solver moves are found matching the same set of user moves, then the tactic lowest in the ordering is used.

5 Player Data

In order to demonstrate the capabilities of our logic grid puzzle tactic labeller, we analyse unlabelled player data collected as part of a previous online user study with methods approved by the Northeastern University IRB.

Players were recruited from the Prolific platform and paid \$2.50 to participate. Each participant was randomly placed into one of nine conditions, which varied in terms of both the puzzles given and the prompts users received during the study. We found that no condition outperformed any other, so in this analysis, we combine all conditions to focus only



* Requesting suggested moves for this augmented state is omitted as no Opening moves are generated.

Figure 3: Example of the oracle and labelling process using the clue “Col. Mustard arrived 2 hours before Ms. Scarlet”. Grey marks are suggested moves, while coloured marks represent the state used to request suggested moves.

on the moves that players made. Players were given up to three logic grid puzzles, of which some were small (two categories of four entities) and some were large (three categories of four entities). Players were able to make either certain or tentative marks on a grid (as shown in Figure 1), and each mark they made was recorded individually.

Participants were not rewarded for correctness and could concede a puzzle at any time, so there was little incentive to cheat (such as by using an automated solver or LLM). Further, available automated solvers are typically specific to a website’s own puzzles, and we did not find an accessible and competent generalized solver. Also, participant behaviour was diverse enough (even among successful users) that we consider it unlikely that LLM solving was prevalent.

6 Results

The study recruited 249 participants (excluding 32 who did not make any marks), and we recorded 507 puzzle attempts of at least one move across all puzzles. Players were permitted to concede a puzzle at any time and either move on to the next puzzle or quit the study. Only 41% of puzzle attempts were successful (resulting in a complete puzzle solution). On average, players spent 3.61 minutes on each puzzle, with

small puzzles taking less time (2.27 min) than large puzzles (5.73 min). Successful and conceding players took around the same time on small puzzles, while successful players took longer (8.28 vs 4.89 min) on large puzzles.

Participants were surveyed on their familiarity with logic puzzles. 69% of players had no experience with logic grid puzzles, although 49% of those players did have experience with other logic puzzles, such as Sudoku. Only 35% of participants had no experience with any kind of logic puzzle.

We calculate results by averaging across all player attempts. Table 1 shows the percent of moves labelled, by player type. Despite the oracle suggesting moves without consulting the solution, tactic labelling was most successful for moves that were correct with respect to the solution. Correct moves were labelled with a tactic an average of 65% of the time, whereas incorrect moves were only labelled 18% of the time. On average, more moves were correct than incorrect (58% vs 28%), with around 24% of moves considered neutral (tentative and erase marks).

We compared the labelling of moves in successful and conceding players. Successful players’ correct moves were labelled more frequently (77%) compared to conceding players’ (56%). On the other hand, incorrect moves were

Player	Correct	Incorrect	Overall
Online (overall)	65%	18%	48%
Online (successful)	77%	14%	64%
Online (conceding)	56%	19%	37%
Expert*	97%	0%**	97%
Random Probabilistic*	48%	41%	46%
Random Correct*	65%	N/A**	65%

* The expert and random correct agent were always successful; the probabilistically random agent always conceded.

** The expert made nearly 0% incorrect moves on average, and the random correct agent made no incorrect moves, by design.

Table 1: Percentages of moves labelled by player type, averaged over all player attempts. All player groups were found (by a Games-Howell pairwise test) to be statistically different ($p < .01$) with respect to the percentage of correct moves labelled.

labelled about the same percent of the time for both groups. On average, successful players were more likely to make correct moves (74% vs 47%).

6.1 Tactic Agreement

We compared tactics labelled in different successful attempts on the same puzzle (“agreement”). We calculate agreement for each puzzle by finding the percentage of successful player attempts using each tactic, and we average the percentages for tactics that are labelled in at least one successful attempt. We found that there is a moderate degree of agreement between successful players; on average, if a given tactic is labelled in one successful player’s attempt, it is labelled in 67% of all successful attempts on that puzzle.

Comparing successful and conceding attempts, tactics were more likely to be labelled in successful attempts than conceding. For 9 of 15 tactics labelled in at least one player attempt, successful players were significantly more likely to make a move labelled with the tactic if it was present in the oracle’s suggested moves (Table 2). The other tactics were labelled less than 5% of the time that they were suggested, for both successful and conceding players. Only “Trans-False” was labelled significantly more often for conceding players, but as it is labelled very rarely for all players and is one of the most complex tactics, this can likely be explained by labelling error.

In general, more complex tactics are more likely to be labelled, as shown in Table 2. However, the complexity of tactics is determined subjectively, and some user moves could be labelled by multiple tactics making the same mark. To analyse the effect of the subjective ordering on tactic labelling, we calculated how often each pair of tactics could both be used to explain a user’s move. Most pairs were seldom or never seen together, and those that were more frequently seen together are generally clustered at the high end of the subjective ordering, suggesting that the ordering of the most complex tactics has the highest effect on the final labels. This is unsurprising, as many of these tactics are closely related logically. The outliers are “Opening” and “Cross-Out”, simple tactics that are frequently paired with

Tactic	Success	Concede	p
Apply-Is	51%	29%	<.01
Cross-Out	26%	17%	<.01
Opening	50%	20%	<.01
Apply-Not	48%	27%	.01
Apply-Or	15%	06%	<.01
Apply-Bef-One	24%	12%	.02
Apply-Bef-N	49%	20%	<.01
Or-Same-Cat	10%	04%	<.01
Bef-Diff-Cat	01%	01%	.949
Bef-No-Info	01%	01%	.119
Bef-N-No-Info	03%	01%	.082
Trans-True	15%	08%	.01
Trans-False	02%	04%	.01
Bef-N-Shift	02%	02%	.95
Bef-N-Cross	<1%	<1%	.95
Trans-Sets	02%	02%	.92

Table 2: Differences in tactic choices between successful (Success) and conceding (Concede) online players. Values indicate the likelihood of the tactic being labelled when it was in the oracle’s suggested moves. Bolded values indicate a statistically significant difference ($p < .05$) between Solved and Conceded, determined by Welch’s t-test with false discovery control by the Benjamini-Hochberg method. The tactics are ordered from least to most complex, according to the tactic ordering. Values are averaged over all player attempts.

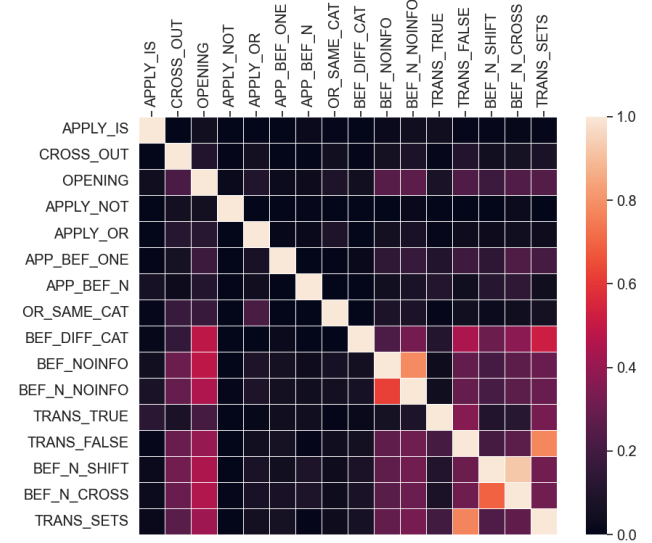


Figure 4: How frequently user moves could simultaneously be labelled as multiple tactics. Values represent the percentage of the time that the column tactic was possible over all times that the row tactic was possible.

the most complex tactics. This can be explained by the fact that “Opening” and “Cross-Out” are based on global constraints that apply to many different grid states. The heatmap in Figure 4 shows all pair frequencies.

6.2 Comparison to Expert and Simulated Data

For comparison to the player data, we created two additional datasets. To generate “expert data”, the first author solved each of the puzzles used and labelled their moves with the tactics they used, so the labeller may be evaluated on ground truth data. Second, two random agents also played each of the puzzles (“simulated data”).

Comparison to Expert Data To evaluate the labeller under near-ideal conditions, we tested the tactic labeller on annotated playtraces of the first author, an expert solver with knowledge of the tactics known by the oracle who made correct moves an average of nearly 100% of the time, with vanishingly few incorrect moves. On average 97% of their moves were labelled. However, only 83% of their moves were labelled correctly with respect to their annotations, with another 14% that were assigned a label inconsistent with their annotation. Of those assigned an incorrect label, 60% were assigned a label higher in the tactic ordering, despite the tactic labeller preferring lower tactics.

Comparison to Simulated Data We tested the tactic labeller on playtraces generated by two random agents, the probabilistically random agent and the random correct agent, each of which was run 20 times per puzzle. The probabilistically random agent makes marks according to their frequency in the solutions to the tested puzzles. At each turn until the grid is completely filled, it selects a random unfilled spot on the grid and fills it with an “O” 25% of the time and an “X” the other 75% of the time. The random correct agent also selects a random unfilled grid cell on each turn, which it fills with the mark indicated by the solution (whether “O” or “X”), until the puzzle is solved.

The probabilistically random agent makes correct moves on average 63% of the time, and on average 48% of correct moves are labelled. The random correct agent’s moves are labelled an average of 65% of the time.

7 Discussion

Although it is impossible to say exactly why players make the moves they do without their explicit explanations, we have demonstrated a relatively simple process that shows promise in annotating moves with players’ possible thought processes. In particular, on successful attempts, most (77%) correct moves are aligned to an existing solver tactic, and attempts for the same puzzle show a moderate (67%) amount of agreement on which tactics are included. That suggests that there is some commonality to the solution process that is shared between players and partially captured by the tactics we have labelled.

Examining the comparison data, the first author’s expert moves are correctly labelled 83% of the time, suggesting that the labeller is able to successfully capture tactics when the player and oracle are aware of the same ones. Surprisingly, of the expert moves assigned an inconsistent label, 60% are assigned a label higher in the tactic ordering. This shows that the expert did more mental bookkeeping than the augmentation system accounts for (making few explicit deductions).

When the augmentation process does not capture all the deductions that are present in the expert’s mental state, moves are interpreted as more complex moves that skip those deductions.

There is some indication that the tactic labeller captures the skill of the player as well. For the first author, who is aware of all tactics and skilfully applies them, an overwhelming percentage (97%) of moves are labelled. On the other hand, for the study participants, those who were successful saw a moderately high percentage (77%) of correct moves labelled, while conceding players saw little over half (56%) labelled. The random correct agent (65%) falls in between the successful and conceding players, while the probabilistically random agent’s correct moves (48%) are labelled slightly less often than conceding players.

The performance of the random agents compared to the conceding players suggests that conceding players’ correct moves are likely to be guesses (the agents may be considered to represent players that are purely guessing). It also signals that the labeller is less likely to label moves that are guesses. The discrepancy between random agents is likely explained by the fact that the random correct agent is always in a valid grid state, so its moves can appear to follow a pattern that is not there. On the other hand, the probabilistically random agent makes incorrect moves on average 37% of the time and does not follow even the most basic tactics, so its accidental correct moves are unlikely to relate to tactics based on the current grid state. The random correct agent’s moves are likely labelled less often than players’ correct moves for two reasons: first, the agent’s move is less likely to be logical given the current board state; second, for tactics that make multiple marks at once, the agent is less likely to make two marks in a row following the same tactic. The high percentage of correct moves (63%) made by the probabilistically random agent can be explained by the fact that most of a solution is composed of “X” marks, so marking “X” 75% of the time has a high chance of success. This probabilistic behaviour also likely accounts for the fact that the agent’s incorrect moves are labelled twice as often as human players’; by using the correct frequency of marks, it accidentally creates patterns.

Our logic grid puzzle oracle is highly domain specific, but the techniques it uses can be generalized to other deduction puzzles. Suggested moves should have a foundation in human-like reasoning, and designers should account for human error and uncertainty as well, for example by including tentative marks and augmenting the state with moves that could be made through mental bookkeeping, as we have done. The design of our oracle can thus be used as a guide for producing domain-specific oracles for other logic puzzles.

Although the current tactic labeller is far from perfect, it is a first step in understanding players’ reasoning for making individual moves. When improved, such a system would be invaluable in studying the differences between expert and amateur play, as well as identifying where unsuccessful players are struggling. It could also have implications beyond gaming, as it could aid in teaching logical tactics in an educational context. Already, it shows promising results

in detecting the skill level of a player using the percentage of correct moves that are labelled.

7.1 Limitations

There are several notable limitations to this work. Most importantly, without users' explanations of their moves, we cannot determine that their moves actually correspond to the labelled tactics. Also, even with carefully designed tactics, the tactic identification process is highly simplified. We always assign moves to the lowest ranking tactic available, and we only support limited mental bookkeeping. Further, we have not performed any test to ensure that the set of tactics we used is complete, although we are reasonably confident that they fully cover the set of puzzles that can be solved by looping over their clues and applying the tactics relating to one clue at a time (as the solver does). There may be additional puzzles using our clue types that can only be solved by using tactics relating to multiple clues at once; if so, then those puzzles are not solvable using these tactics.

Another limitation of the tactic resolution process is the designer input required. The designer must provide an oracle that suggests moves that are annotated with tactics, and developing such an oracle is a non-trivial task. This is particularly true for games in which the tactics required are not clearly defined or are difficult to identify. Additionally, a tactic ordering must be determined by the designer, in the case that multiple tactics suggest the same move. We found that in logic grid puzzles this is not uncommon, particularly for more complex tactics and those that are closely related.

Our comparison of successful and conceding amateurs with random agents and the first author's expert play suggests that the tactic labeller is only likely to be successful for those playing with some skill. Therefore, for those players that need the most help, the labeller is the least likely to provide an accurate signal of their knowledge. Further, despite the oracle being unaware of puzzle solutions, only a small percent of incorrect moves are labelled across the board, so we are unable to extrapolate *why* a player has made a mistake. Finally, our expert play data itself is limited by the fact that the first author is highly familiar with the labelling and solving process. Although they did not consciously emulate it when they solved the puzzles, such knowledge (and the use of only a single expert player) means the labeller's performance on expert data cannot be generalized.

7.2 Future Work

In the future, we hope to use tactic labelling to identify tactics that players are struggling with and provide tactic-specific support. One way we could do that is by generating minimal "tactic problems" designed to guide users toward a tactic by reducing complexity aside from that tactic, similarly to the work by Jemmali, Seif El-Nasr, and Cooper (2022). This would also help validate our set of tactics, as if our tactics are complete, players would be expected to follow them when presented with a minimal problem.

Our tactic labelling process can be improved in multiple ways. First, we can develop a better understanding of thought patterns by performing "think-out-loud" studies where participants must explain their reasoning for making

moves. Additionally, there are several technical improvements that could be made. We hope to implement Bayesian Knowledge Tracing to give relative confidence to each tactic we assign to a move. We also plan to incorporate information about multiple player moves at once, to better track dependencies between related moves and tactics, as Pelánek (2014) does for Sudoku. We will also study differences between players at different skill levels. Finally, we intend to improve labelling of incorrect moves, possibly by developing a taxonomy of common errors in logical deduction.

Tactic labelling can be used in different areas and domains. We are developing a method of comparing puzzle attempts according to which tactics players use in the process, to see if we can cluster users according to a solution strategy. Finally, in addition to the logic grid puzzle domain, we plan to implement tactic labelling in other puzzle games, such as Sudoku. We also hope to develop a method of tactic discovery that is not dependent on manual analysis. Tactics can be thought of as skipping part of the search space, and we hope to explore search based methods of tactic discovery (and tactic ordering). In particular, we are interested in expanding on work by Browne (2013) on deductive search, Butler, Torlak, and Popović (2017) on Nonograms and Pelánek (2014) on Sudoku.

8 Conclusion

This work introduces a method for labelling player moves with reasoning tactics suggested by a designer-specified oracle. Our system labels the tactic of lowest complexity that corresponds to the move the player made. To demonstrate the power of this simple system, we developed an oracle for logic grid puzzles, a genre of logic puzzle that often makes use of complex reasoning tactics. Rather than simply suggesting correct solution moves, the oracle makes suggestions based on the current grid state — the same information that the player has. In addition, to incorporate player uncertainty and mental bookkeeping, we incorporate marks representing uncertainty and request moves with a limited amount of mental bookkeeping.

We have found that our tactic labelling system is able to apply tactics to over 75% of correct player moves on successful attempts, and the percentage of moves labelled corresponds with player skill. Also, tactics labelled show a moderate agreement between successful attempts of the same puzzle, showing some commonality in solutions. However, we do not have ground truth data to assess the accuracy of the labels. In future work, we intend to address some of the shortcomings of our labelling system by using think-out-loud ground truth data and incorporating Bayesian Knowledge Tracing to analyse users' attempts in more detail, individually and in aggregate.

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